

## Cloud Physics, Fall 2022

### Homework 3

#### Homework Instructions

Due date: 2022/12/28 (Wed.) 10:00 AM

You should submit the following files to NTU COOL before the due date:

- A **pdf file** containing all your answers, including calculations and plots, if required by the question
- **Code(s)** used for calculation and/or plotting (if any)

Late Submission Policy:

- ~~20% deduction between 12/21 10:00 AM ~ 12/28 10:00 AM~~
- **0 points** after 12/28 10:00 AM

Contact TA: b09209005@ntu.edu.tw

This homework contains two problems. Please answer them in the given order.

#### 1. Radar (Chap.11)

(1) Consider the following equations:

$$n(D) = N_0 \cdot D^i \cdot \exp(-\lambda D)$$
$$Z = \int_0^\infty D^6 \cdot n(D) dD$$

Let  $N_0 = 10^4 \text{ (mm}^{-i-1} \text{ m}^{-3}\text{)}$  and  $\lambda = 1 \text{ mm}^{-1}$

Please plot the gamma distribution  $n(D)$  when  $i = -3, 0$ , and  $3$ . For each case, also calculate its radar reflectivity factor ( $Z$ ). Discuss the differences.

[Hint]

- You may refer to page 19 of Chap.11 slides.
- For the plot, you only need to show  $n(D)$  for  $0.1 < D < 20 \text{ mm}$  in the x-axis.

(2) Refer to page 31 of Chap.11 slides and answer the following questions.

- What are the wavelengths of S, C, and X band radars respectively?
- What is the physical mechanism behind a radar bright band (雷達亮帶)? List out two reasons.

## 2. Condensation Growth (Chap.12)

With the given gamma distribution as our initial number density function  $n(D)$ , please calculate the evolution of  $n(D)$  under condensation growth.

Initial number density distribution:

$$n(D) = N_0 \cdot D^3 \cdot \exp(-\lambda D)$$

$$N_0 = 10^6 \mu m^{-4} m^{-3}; \lambda = 1 \mu m^{-1}$$

Consider the condensation growth of a single drop:

$$\frac{dm}{dt} = 4\pi r f_v \cdot \frac{\Delta S}{A_d + A_k}$$

where  $f_v = 1$  (for cloud drops), and  $\frac{\Delta S}{A_d + A_k} \approx 0.01$ .

Hence for the population (mode) of cloud drops:

$$\frac{dQ}{dt} = \int \frac{dm}{dt} \cdot n(D) dD \approx \frac{2\pi\Delta S}{A_d + A_k} \int D \cdot n(D) dD$$

(1) Integrate the condensation process for 10 minutes, and plot the number density function at **time = 0 min, 5 min, and 10 min.**

(2) Plot the evolution of  $M_2$  and  $M_3$  with time.

[Hint]

- You may refer to page 25 and pages 32~33 of Chap.12 slides. Not all equations needed are listed here.
- For the plot, you only need to show  $n(D)$  for  $0.1 < D < 15 \mu m$  in the x-axis.